Data Cleaning Change Log

DecodeLabs Data Analytics | Project 1: Data Cleaning & Preparation | Prepared by Favour Daniel

1. Dataset overview

The raw file contained 1,200 order records across 14 columns (OrderID, Date, CustomerID, Product, Quantity, UnitPrice, ShippingAddress, PaymentMethod, OrderStatus, TrackingNumber, ItemsInCart, CouponCode, ReferralSource, TotalPrice), covering orders from 2023-01-01 to 2025-06-30. Every column was checked for missing values, duplicates, inconsistent text, wrong data types and logical errors. The cleaned file is Cleaned_Dataset.xlsx.

2. Audit findings

Check	What was tested	Finding
Missing values	Null and blank cells in all 14 columns	Only CouponCode had gaps: 309 of 1,200 records (25.75%). All other columns were complete.
Duplicates	Full-row duplicates, duplicate OrderID, duplicate TrackingNumber, and repeats of the same customer, date, product, quantity and price	None found. All 1,200 OrderIDs and TrackingNumbers are unique.
Date formats	Data type and format of every value in Date	All 1,200 are true Excel dates, already in YYYY-MM-DD, with no text dates and no time parts.
Text consistency	Leading and trailing spaces, double spaces, case, spelling variants in Product, PaymentMethod, OrderStatus, ReferralSource, CouponCode	No variants, no stray spaces, proper case throughout.
ID patterns	OrderID (ORD + 6 digits), CustomerID (C + 5 digits), TrackingNumber (TRK + 8 digits), ShippingAddress	All values match their expected pattern.
Numeric logic	Quantity and UnitPrice above zero; TotalPrice = Quantity x UnitPrice; ItemsInCart not below Quantity; price outliers	No errors. Totals reconcile on every row and no outliers beyond 3 standard deviations.
Numeric precision	TotalPrice and UnitPrice stored to 2 decimals	29 TotalPrice values carried floating-point noise beyond 2 decimals (largest difference about 0.0000000000001). Cells also used General format.

3. Change log

Change ID	Description	Impact	Status
CR001	Filled 309 blank CouponCode values with "No Coupon" instead of using the mode or deleting the rows. A blank here means the customer used no coupon, so the mode (FREESHIP) would have invented discounts that never happened.	Preserved all 309 records. Coupon column is now analysable (FREESHIP 313, No Coupon 309, WINTER15 292, SAVE10 286).	Resolved
CR002	Rounded TotalPrice and UnitPrice to 2 decimals and applied a 0.00 number format.	Removed floating-point noise on 29 TotalPrice values. All prices now display consistently, e.g. 361.00 instead of 361.	Resolved
CR003	Applied the yyyy-mm-dd format explicitly to the Date column and confirmed every value is a true date.	No values changed. Format locked to ISO 8601 and verified at 0 errors.	Resolved

Change ID	Description	Impact	Status
CR004	Trimmed whitespace on all text columns as a safeguard.	0 values changed, since none had stray spaces. Kept so the process is repeatable.	Resolved
CR005	Duplicate removal run on OrderID, TrackingNumber and full rows.	0 duplicates found, so 0 rows removed. Record count stays at 1,200.	Resolved
CR006	Formatted the sheet for use: header styling, frozen header row, column filters and sensible column widths. Added a Verification sheet with live formulas.	No data changed. Makes the file ready for the next project and lets anyone re-check the results.	Resolved

4. Verification gate (Project 2 threshold)

The brief requires proof of zero duplicate IDs and zero incorrectly formatted dates. These checks are built as live formulas in the Verification sheet of Cleaned_Dataset.xlsx and all passed:

Check	Result	Expected	Status
Total records	1,200	1,200	PASS
Duplicate OrderIDs	0	0	PASS
Duplicate TrackingNumbers	0	0	PASS
Dates not stored as real dates	0	0	PASS
Blank dates	0	0	PASS
Dates outside 2023-01-01 to 2025-06-30	0	0	PASS
Blank cells anywhere in the dataset	0	0	PASS
TotalPrice not equal to Quantity x UnitPrice	0	0	PASS

5. Notes and assumptions

Several customers appear more than once (22 rows across 11 customers) with different orders and addresses. These are repeat purchases, not duplicates, because each has its own OrderID and TrackingNumber, so they were kept. Tools used: Python (pandas) for the audit and Excel for the final file.